

Predicting Customer Subscription Status Using Machine Learning

Motivation

Online retail platforms are increasingly relying on data to understand customer behavior and improve long-term customer engagement. One important business question is identifying which customers are likely to subscribe to a service, such as a loyalty or premium membership program. Predicting subscription status enables companies to target their marketing efforts more effectively, personalize shopping experiences, and utilize their resources more efficiently.

I chose this problem because shopping behavior data is realistic and closely connected to everyday experiences. As a consumer who frequently shops online, I am interested in how companies analyze customer activity to make predictions and strategic decisions. This problem allowed me to work with data that reflects real-world purchasing patterns while applying ML techniques learned in class. Additionally, subscription prediction is a practical classification task that highlights the importance of data preprocessing, model comparison, and evaluation metrics when building reliable predictive systems.

Problem Translation

The real-world problem of understanding customer subscription behavior is a binary classification task. Each customer is labeled as either subscribed (“YES”) or not subscribed (“NO”).

Machine Learning Task:

Predict whether a customer has an active subscription based on their demographic information, purchasing behavior, and shopping preferences.

Modeling Goal:

To train and compare multiple supervised learning models to accurately classify customers’ subscription status and identify which models perform best.

Data Description

The open source Kaggle dataset I used in this project is a Shopping Behavior & Preferences dataset, which captures customer interactions with products and purchasing patterns.

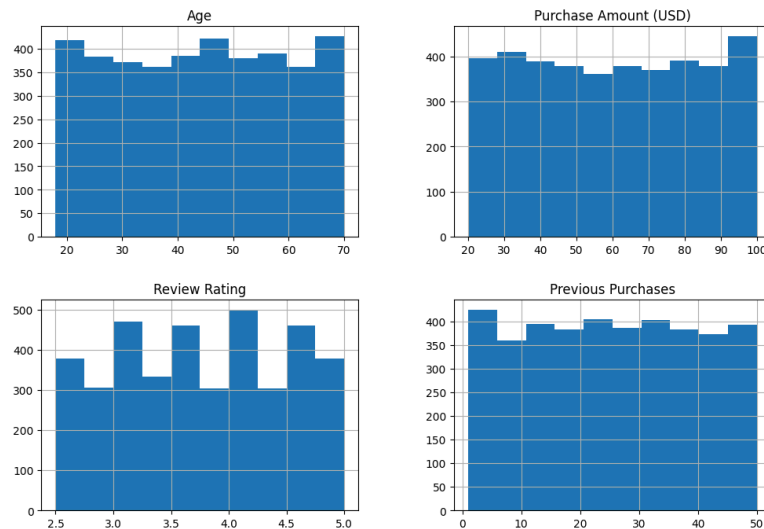
<https://www.kaggle.com/datasets/ranaghulamnabi/shopping-behavior-and-preferences-study>

Key Features:

- Demographics: Age, Gender, Location
- Purchasing Behavior: Purchase Amount, Previous Purchases, Frequency of Purchases
- Product Preferences: Item Purchased, Category, Size, Color, Season
- Transaction Details: Payment Method, Shipping Type, Discounts, Promo codes
- Target Variable: Subscription States (Yes or No)

The dataset contains both numerical and categorical variables

Distributions of Numerical Features

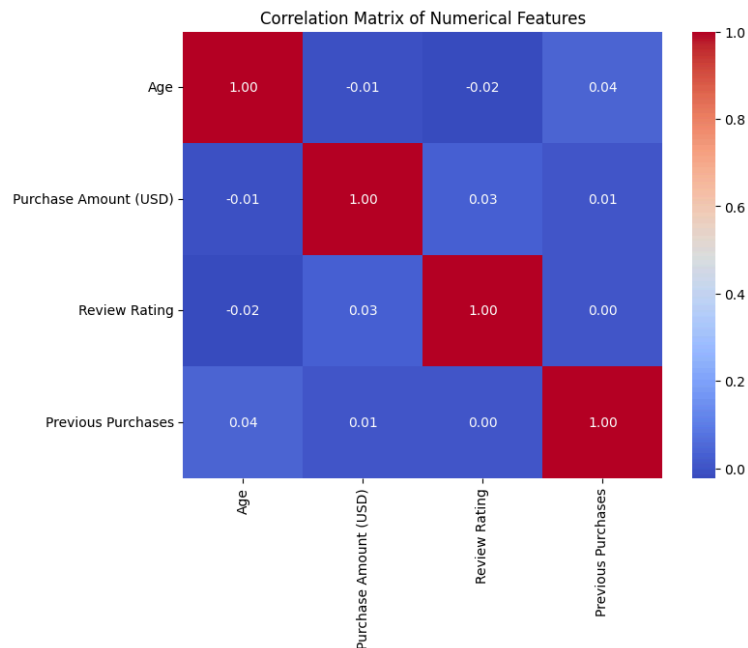


Exploratory Data Analysis

To better understand the structure/quality of the data

- Numerical features such as age, purchase amount, and previous purchases showed reasonable ranges with no extremes.
- The subscription status was moderately imbalanced but still suitable for classification.

Bar plots revealed differences in subscription rates across purchase frequency and discount usage



Heatmaps were used to visualize the correlations between numerical features.

Data Quality

- There were no major missing values
- The variables were well defined with consistent labels
- Not many outliers, and didn't require removal

Data Preparation

Feature and Target Definition

- Target Variable: Subscription Status
- Input Features: All remaining variables except Customer ID

Preprocessing Steps

- Numerical features were passed through without modification

- Categorical features were encoded using OneHotEncoding to allow models to process non-numeric data
- A ColumnTransformer was used to apply preprocessing consistently across all models
- Pipelines ensured that preprocessing and model training occurred together, which prevented data leakage.

Selected Models

Six machine learning models were selected

- (1) Logistic Regression
- (2) K-Nearest Neighbors
- (3) Decision Tree
- (4) Random Forest
- (5) Support Vector Machine
- (6) Naive Bayes

These models provided a diverse comparison between linear and nonlinear approaches.

Evaluation Metrics

Metrics used to evaluate were:

- Accuracy: Overall correctness
- Precision: Proportion of predicted subscribers that are correct
- Recall: Identifies actual subscribers
- F1-Score: Balance between precision and recall
 - Emphasized because the subscription prediction benefits from balancing false positives and negatives

I explicitly defined the positive class as “Yes” to ensure correct metric calculation.

Hyperparameter Optimization

Models Requiring HPO

- KNN
- Decision Tree
- Random Forest
- SVM

Applied HPO to models that benefited the most from it:

- KNN: Number of neighbors
- Decision Tree: Max depth
- Random Forest: Number of trees, max depth
- SVM: Regularization parameter and kernel type

Final configurations were selected based on validation performance. Reduced computational cost while maintaining strong performance

Model Evaluation Tables:

The top three models were evaluated on the test set to assess the real-world performance. Results were consistent with validation findings, indicating minimal overfitting.

Train Results

	Accuracy	Precision	Recall	F1
Model				
Logistic Regression	0.858608	0.687300	0.873813	0.769415
KNN	0.788645	0.702020	0.377205	0.490733
Decision Tree	0.913187	0.794811	0.914518	0.850473
Random Forest	0.878022	0.688785	1.000000	0.815717
SVM	0.854945	0.666993	0.924016	0.774744
Naive Bayes	0.839194	0.626701	1.000000	0.770518

Validation Results

	Accuracy	Precision	Recall	F1
Model				
Logistic Regression	0.813675	0.621891	0.791139	0.696379
KNN	0.705983	0.412500	0.208861	0.277311
Decision Tree	0.825641	0.647368	0.778481	0.706897
Random Forest	0.832479	0.617188	1.000000	0.763285
SVM	0.815385	0.613636	0.854430	0.714286
Naive Bayes	0.832479	0.617188	1.000000	0.763285

Model	Accuracy	Precision	Recall	F1
SVM	High	High	Very High	Best
Naive Bayes	High	High	Very High	Best
Random Forest	High	Moderate	High	Strong
Logistic Regression	Moderate	Moderate	Moderate	Stable
Decision Tree	Moderate	Moderate	Moderate	Acceptable
KNN	Lowest	Lower	Lower	Weakest

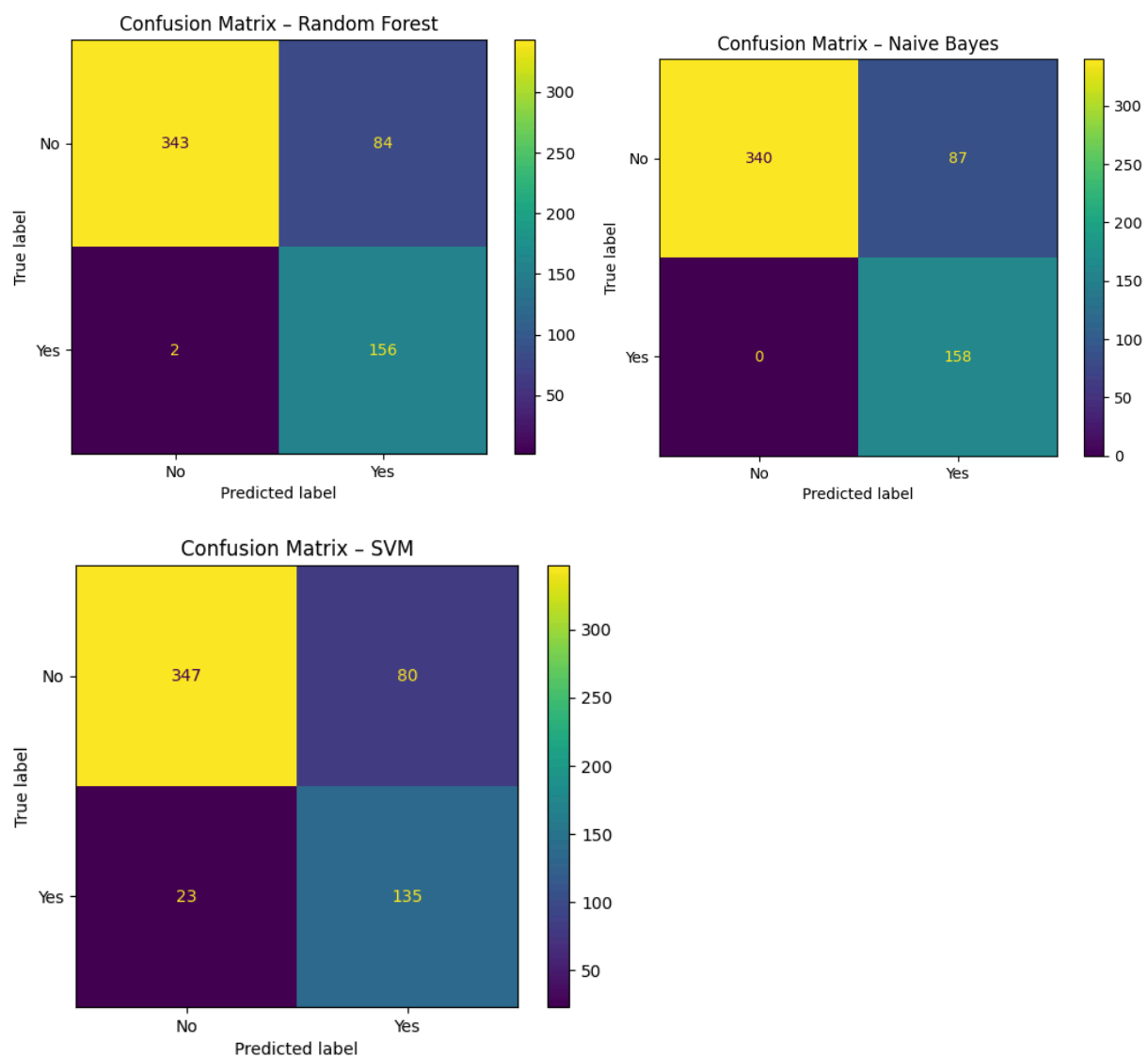
All models were evaluated on a validation set using the selected metrics

Observations:

- SVM and Naive Bayes achieved the highest F1-Scores
- Random Forest performed consistently well across all metrics
- Logistic Regression served as a reliable baseline
- KNN and Decision Tree showed slightly lower performance

Top 3 Models Selected:

- (1) SVM
- (2) Naive Bayes
- (3) Random Forest



Model Predictions and Interpretation

To assess the performance of the top 3 models, I generated predictions on the test dataset and visualized the results using confusion matrices. These matrices illustrate the model's ability to correctly classify customers who are likely to subscribe to the service vs. those who are not

Random Forest

- True Positives (Yes, correctly predicted): 156
- False Negatives (Yes predicted as No): 2
- False Positives (No predicted as Yes): 84
- True Negatives (No correctly predicted): 343

The Random Forest model demonstrates strong predictive power, particularly for identifying subscribers. It maintains a high recall for the "Yes" class while keeping a reasonable precision for the "No" class. Most errors occur in misclassifying some "No" customers as "Yes."

Naive Bayes

- True Positives: 158
- False Negatives: 0
- False Positives: 87
- True Negatives: 340

Naive Bayes achieves perfect recall for the "Yes" class, capturing all potential subscribers. However, it slightly over-predicts subscription status, leading to more false positives. A trade-off is acceptable when missing potential subscribers is more costly than mistakenly targeting non-subscribers.

SVM

- True Positives: 135
- False Negatives: 23
- False Positives: 80
- True Negatives: 347

The SVM model performs well for the "No" class but struggles a little with the "Yes" class, resulting in a higher number of false negatives. This indicates that SVM is more conservative in the prediction of subscribers and may miss potential customers who could benefit from the targeted marketing.

- Across all the models, Random Forest offers the best balance of precision and recall for predicting subscription status. Naive Bayes maximizes recall at the cost of more false positives, while SVM favors precision for the "No" class. These observations highlight the importance of selecting a model based on the goal of the business, whether it is minimizing missed subscribers or avoiding unnecessary promotions.
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Conclusion

This project was to predict customer subscription status for an online platform. Using the Kaggle dataset I found, I processed and prepared the data, trained six machine learning models, tuned their hyperparameters, and evaluated their performance.

Key Findings:

- Random Forest provided the strongest overall performance, balancing precision and recall effectively.
- Naive Bayes excelled at identifying all potential subscribers but increased false positives.
- SVM was more conservative and had a lower recall for the subscriber class.

Predicting subscription status allows businesses to target marketing efforts more efficiently and personalize customer experience. Using a well-performing model like Random Forest ensures resources as allocated to customers most likely to engage, improving both profitability and customer satisfaction